

How to install OpenSSH on Windows

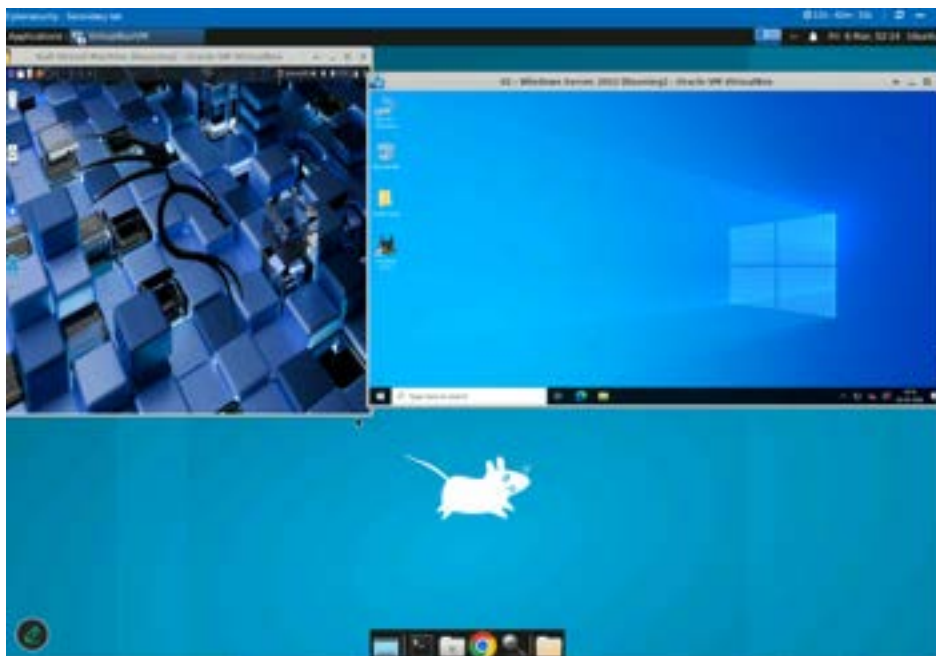
Step One-

Open Virtual Machine waits for the page to load.

Log in with the credentials

Labuser

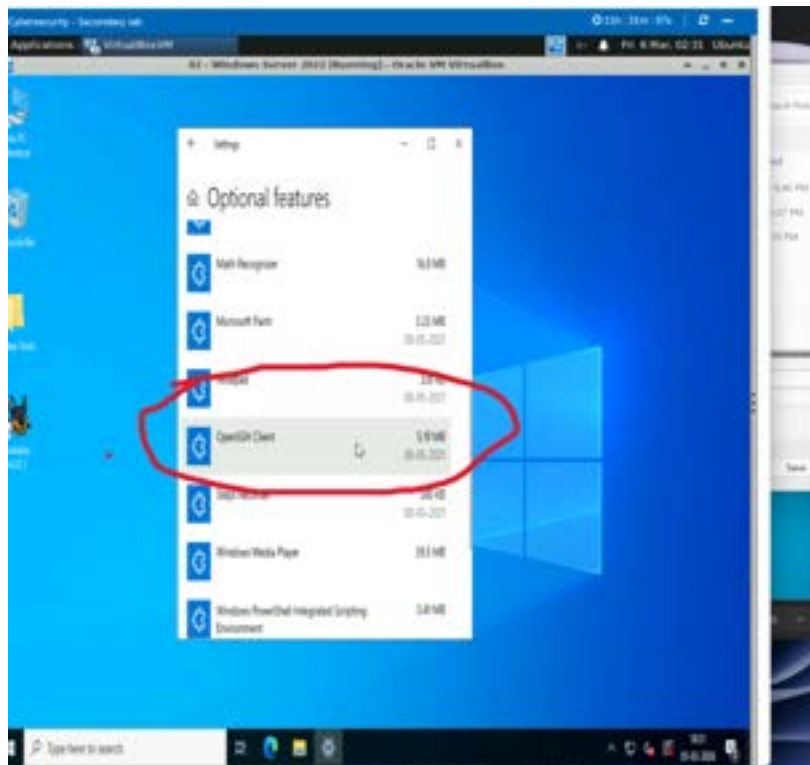
Password.1!!



Step Two-

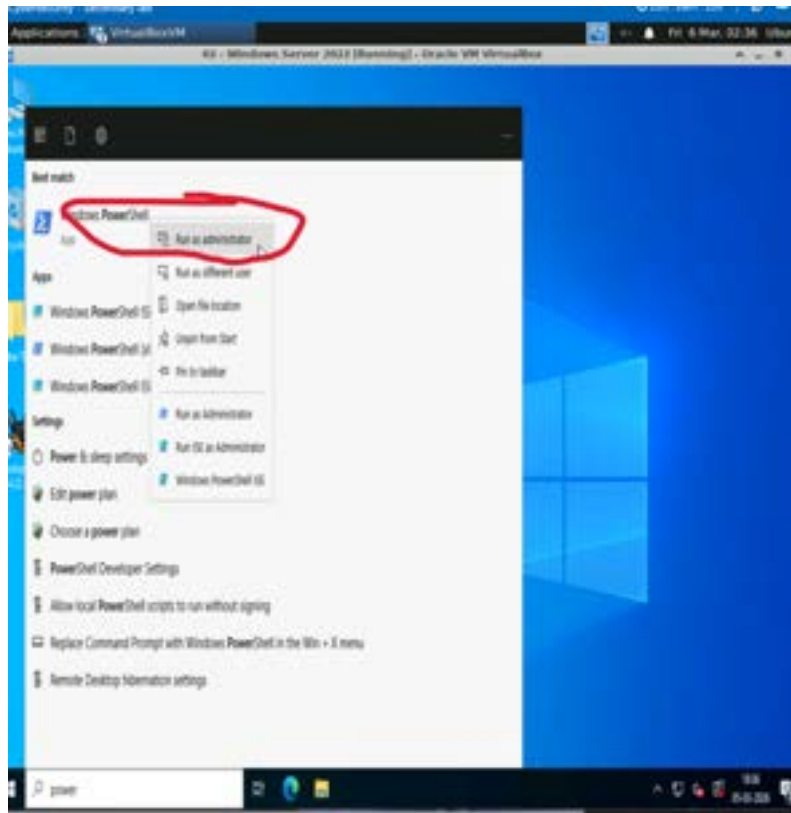
Windows Server open Settings > Apps > Optional Features > Add a feature

Select OpenSSH Server and install it



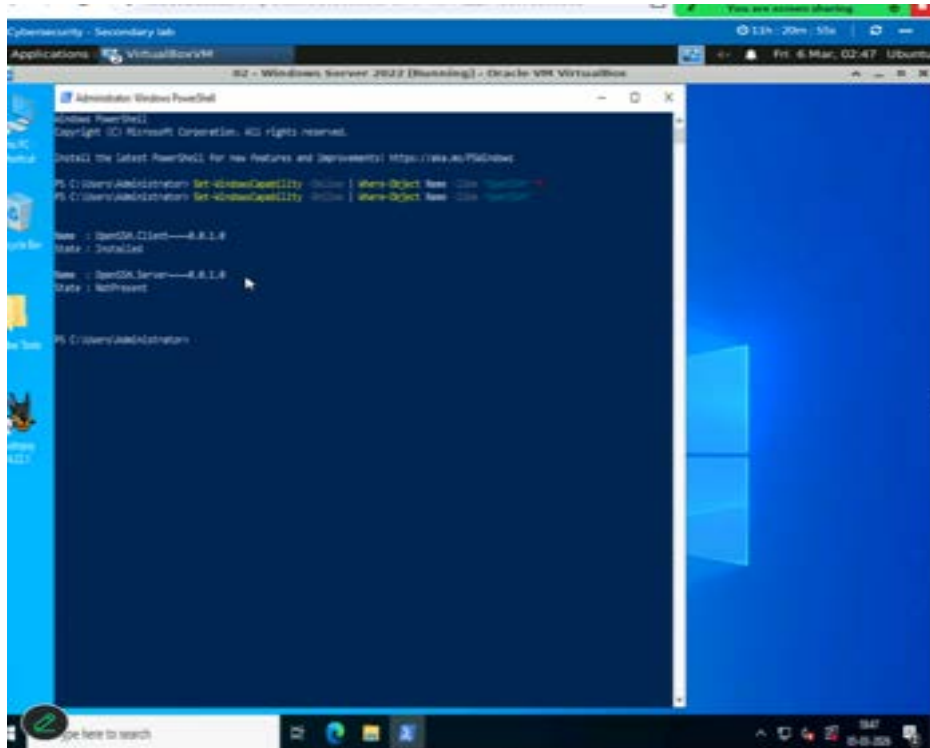
Step Three-

Open Powershell as an administrator and run the following:



Step Four-

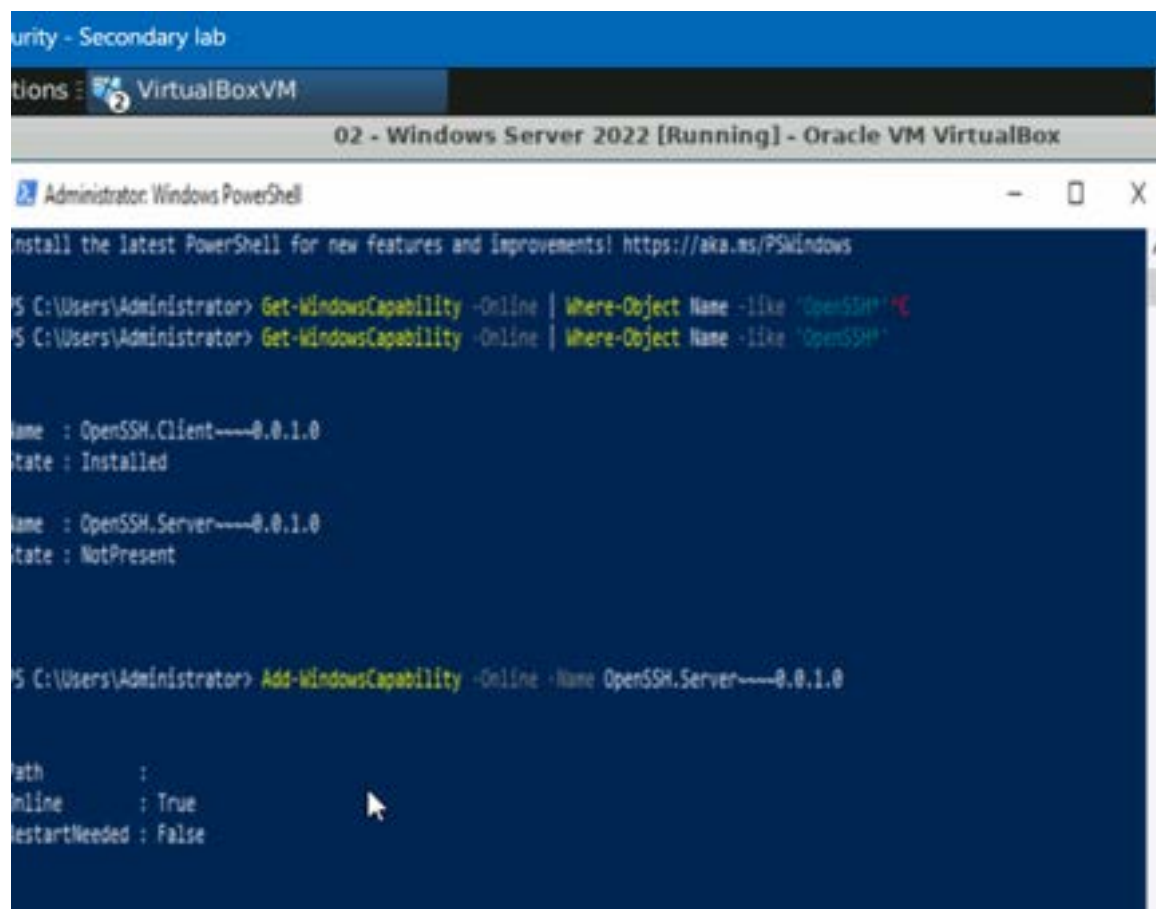
Run the command- Get-WindowsCapability -Online | Where-Object Name -like 'OpenSSH*'



This Lists all OpenSSH features available (Client and Server). (Picture Above)

Step Five-

Run the command- **Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.1.0**



```
Security - Secondary lab
VirtualBoxVM
02 - Windows Server 2022 [Running] - Oracle VM VirtualBox
Administrator: Windows PowerShell

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Get-WindowsCapability -Online | Where-Object Name -like 'OpenSSH*'
PS C:\Users\Administrator> Get-WindowsCapability -Online | Where-Object Name -like 'OpenSSH*'

Name : OpenSSH.Client~~~~0.0.1.0
State : Installed

Name : OpenSSH.Server~~~~0.0.1.0
State : NotPresent

PS C:\Users\Administrator> Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.1.0

Path :
Online : True
RestartNeeded : False
```

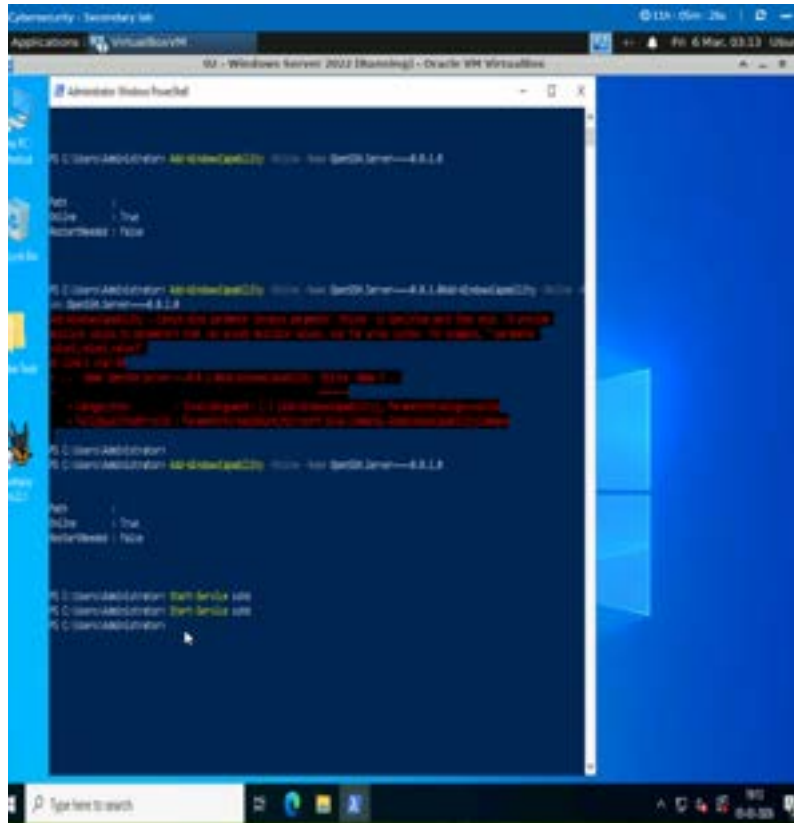
(Installs the OpenSSH Server component)

Step Six-

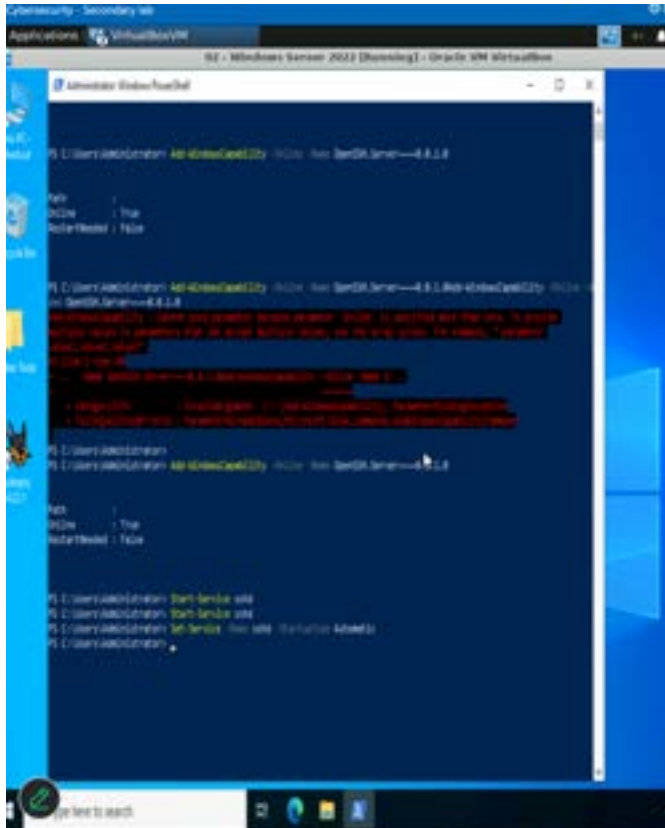
Start and enable the OpenSSH Service

Run the following powershell commands to start the service and make it always run on startup

Run the command- Start-Service sshd



Run Command-Set-Service -Name sshd -StartupType Automatic

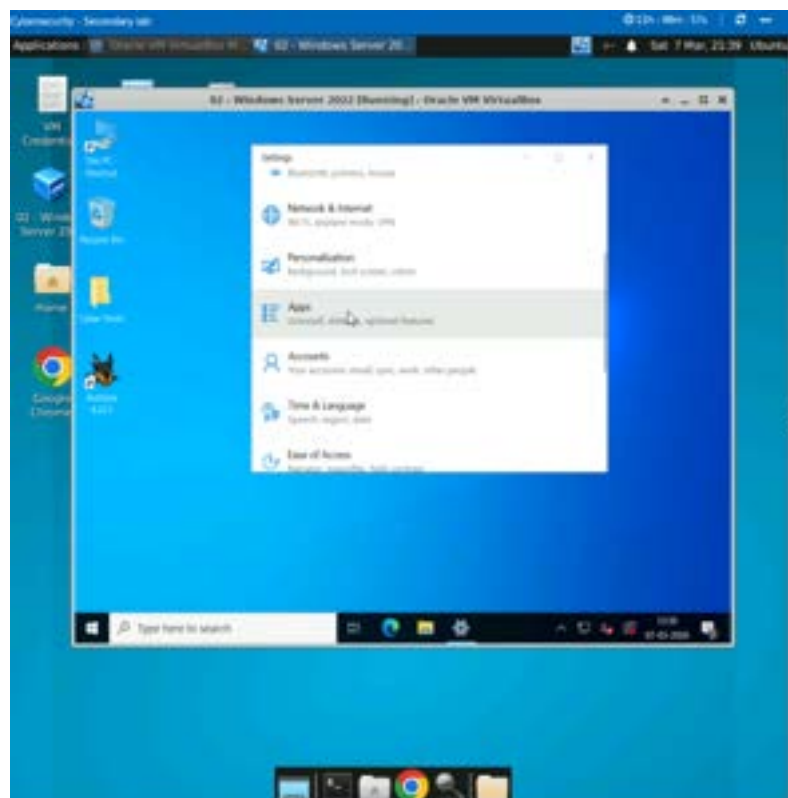


How to configure logging (Forum Post)

Logging and Viewing Remote SSH Connections on Windows (OpenSSH Server)

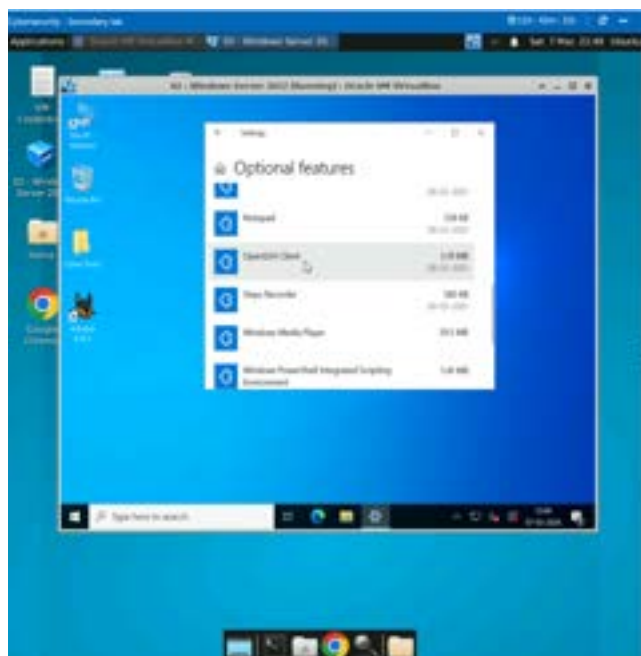
Step 1: Install OpenSSH Server

1. Open Settings → Apps → Optional features





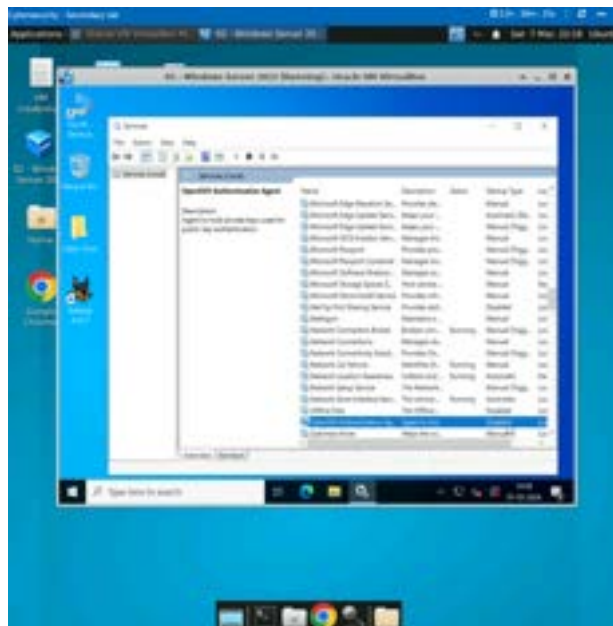
2. Scroll down to check if OpenSSH Server is listed.



If not, click Add a feature, search for OpenSSH Server, and install it.

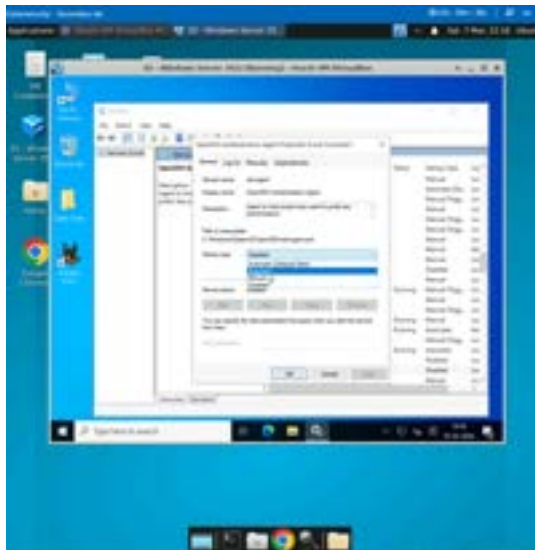
Open Services (Win + R → type services.msc).

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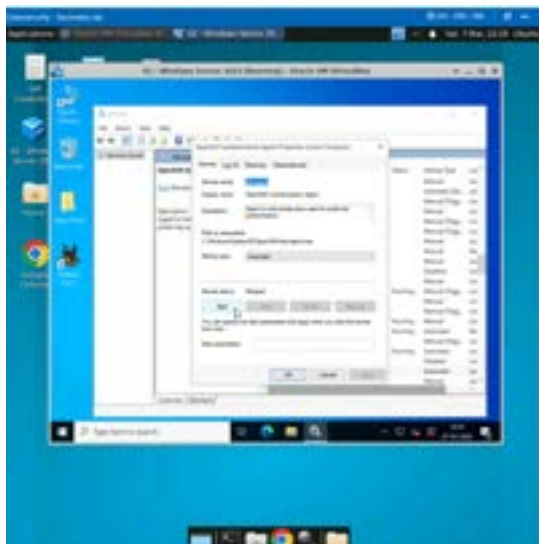


Right Click, then select Properties

Set Startup type: (Automatic)



Click Start if the service is not already running.



Step 3: Verify Logging is Enabled

Open PowerShell as Administrator.

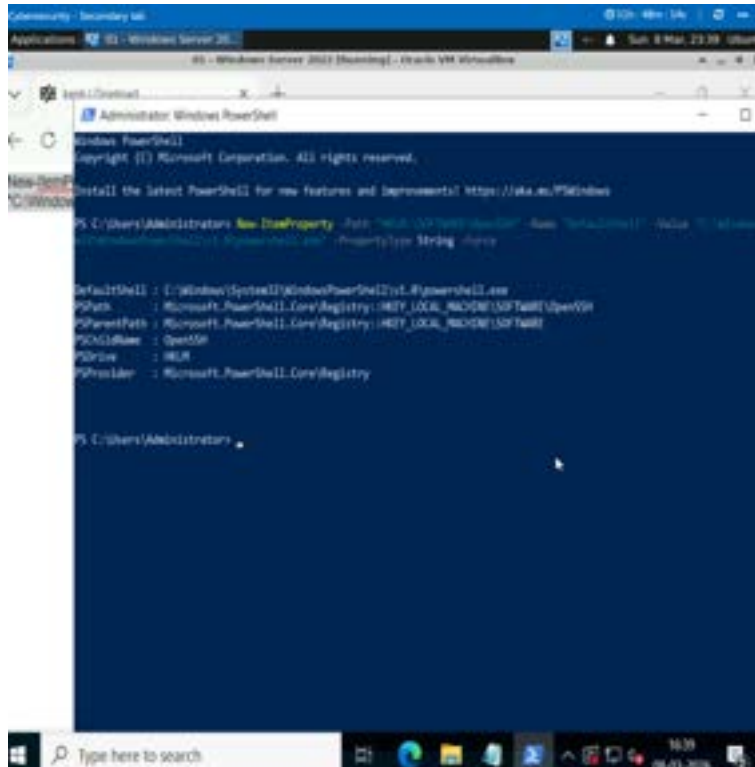


Run the following command:

```
New-ItemProperty -Path "HKLM:\SOFTWARE\OpenSSH" -Name "DefaultShell" -Value
```

```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -PropertyType String -Force
```

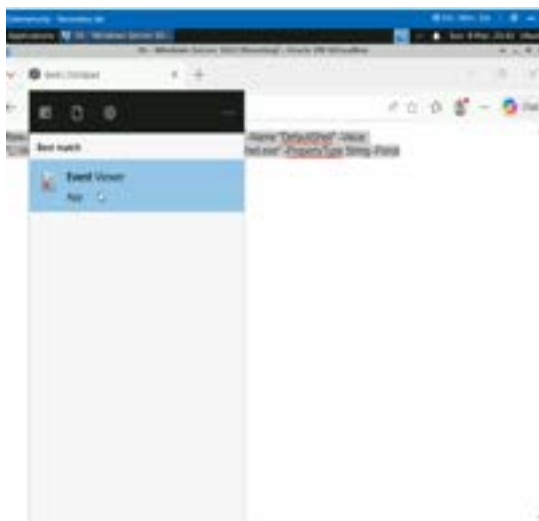
To copy and paste command Use dontpad.com to transfer text over to Virtual Machine on Windows Server



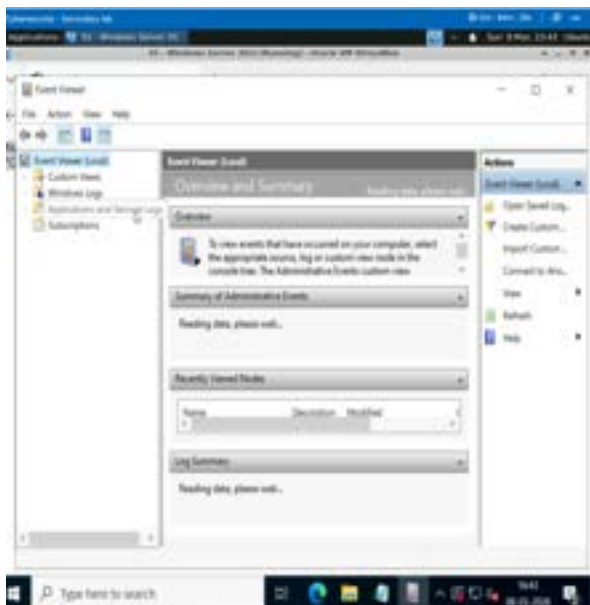
Logging is automatically written to Event Viewer.

Step 4: View SSH Logs in Event Viewer

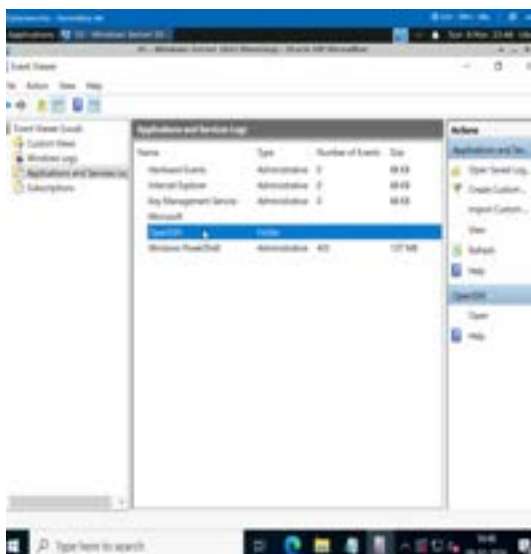
Press Win + R → type eventvwr.msc.



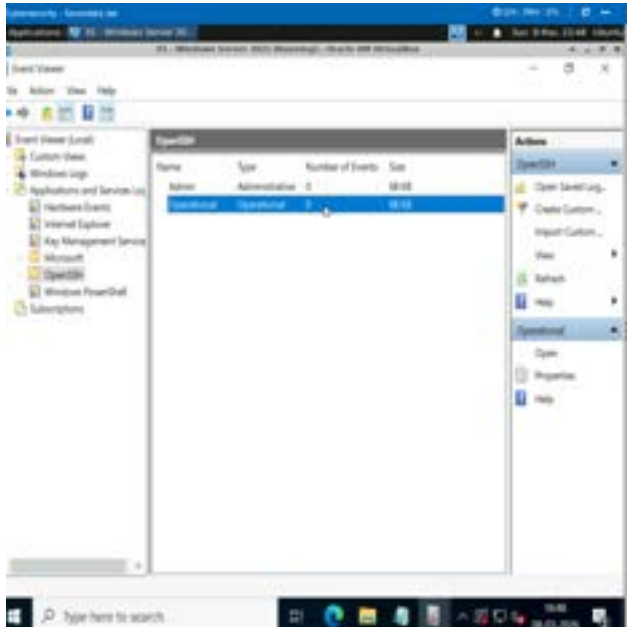
Expand: Applications and Services Logs



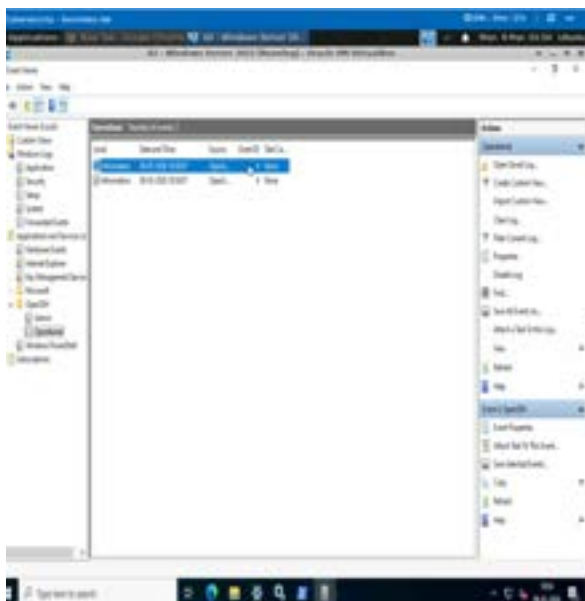
OpenSSH



Operational

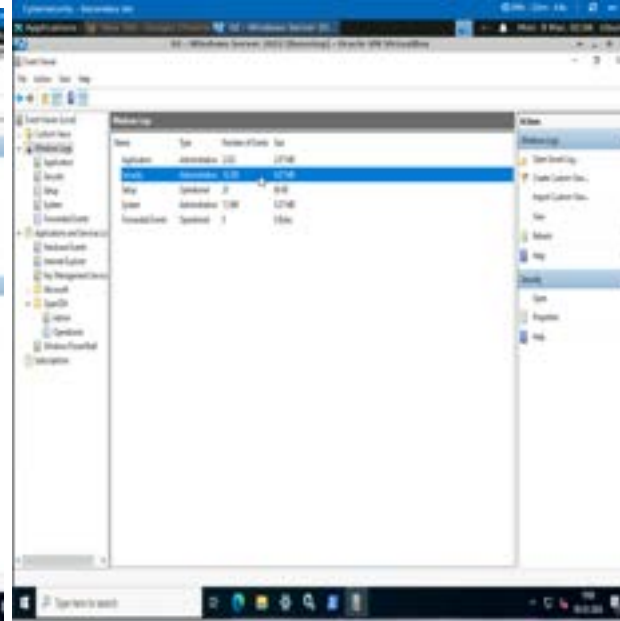


- Event ID 4 = login attempt

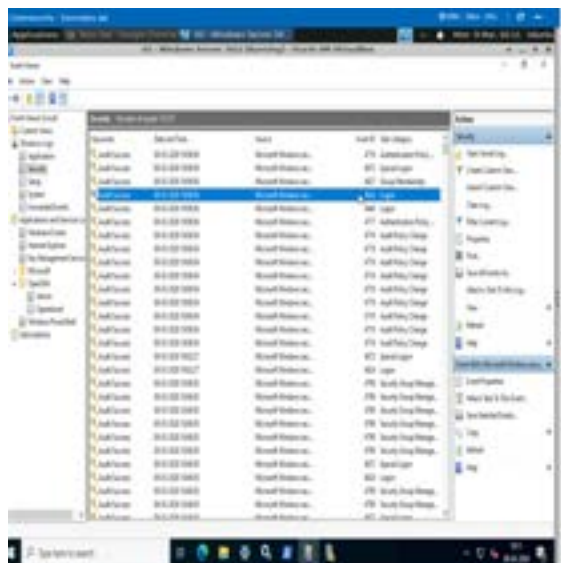


If **Event ID 10 isn't showing**, that's actually **normal on many Windows systems**. Newer Windows OpenSSH versions **don't always generate Event ID 10 anymore**, even though older guides still mention it.

Check Windows Logs → Security:



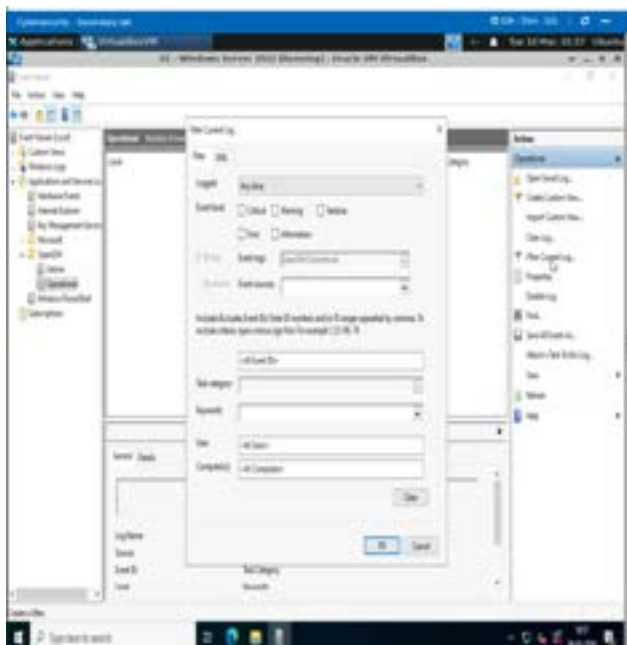
Event ID 4624 = successful logon



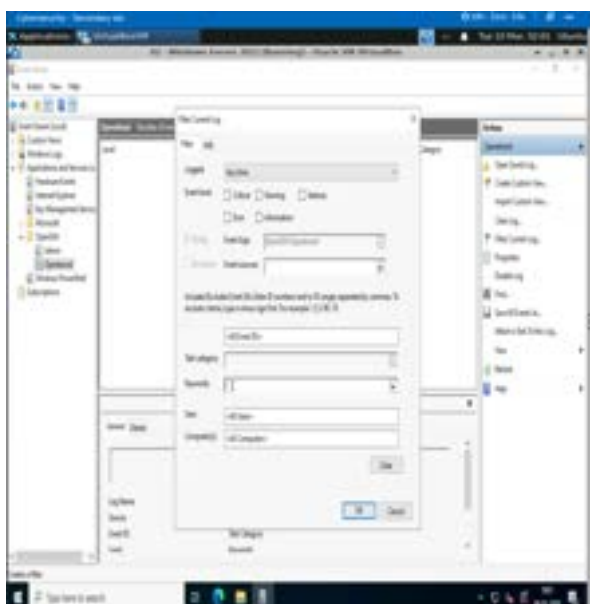
Step 5: Filter Logs for SSH Activity

In Event Viewer, right-click the OpenSSH Operational log

Select Filter Current Log...

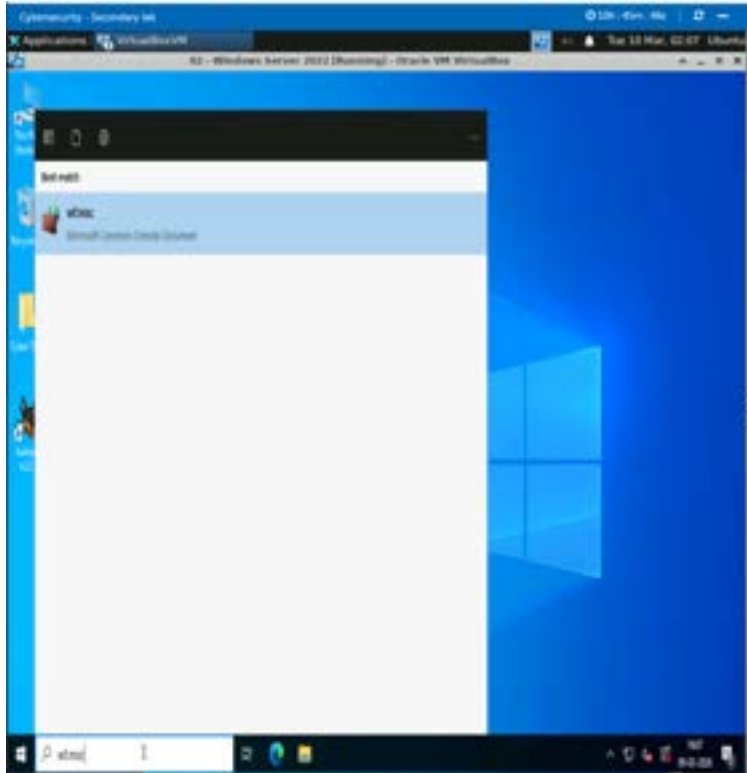


Enter keywords like 'sshd' or filter by Event IDs (4, 10)



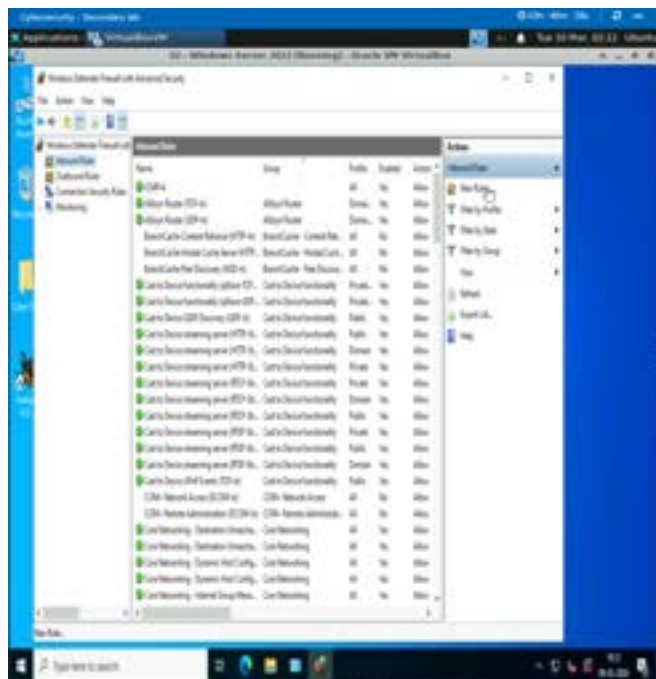
How to configure Windows Firewall rules

Step 1: Open Windows Defender Firewall

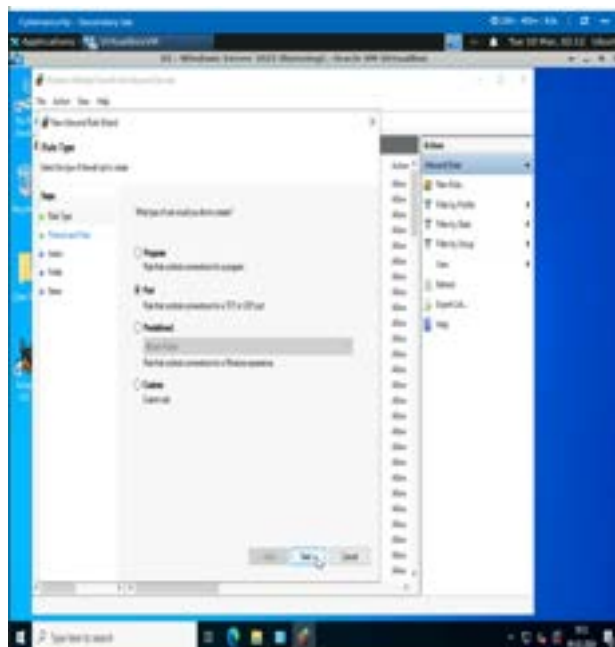


This opens the Windows Defender Firewall with an Advanced Security console.

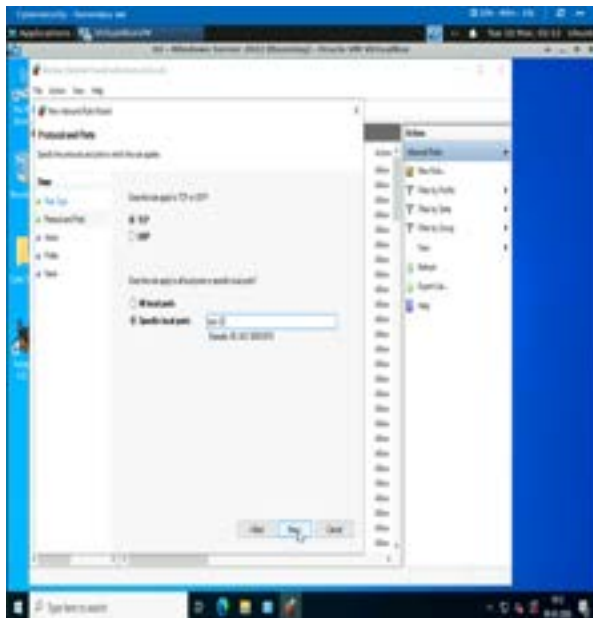
In the right panel, click New Rule...



Select Port → click Next.

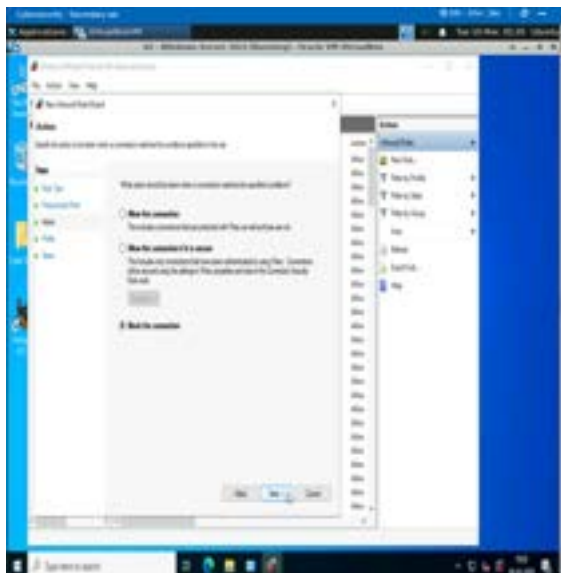


Choose TCP and type 22 (default SSH port) → click Next

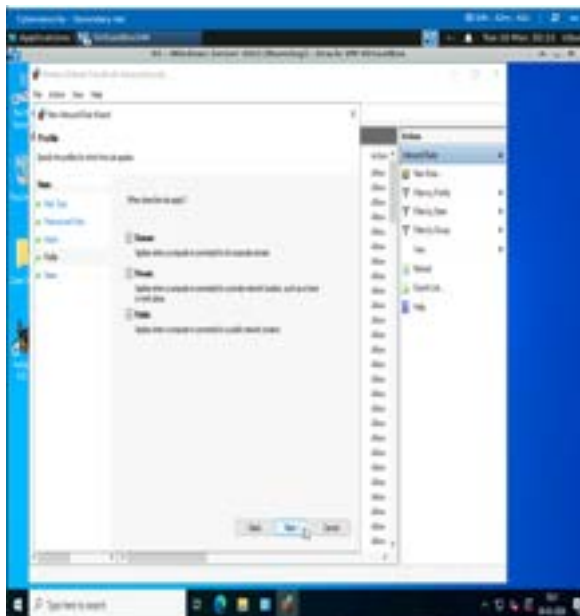


Step 3: Block the Connection

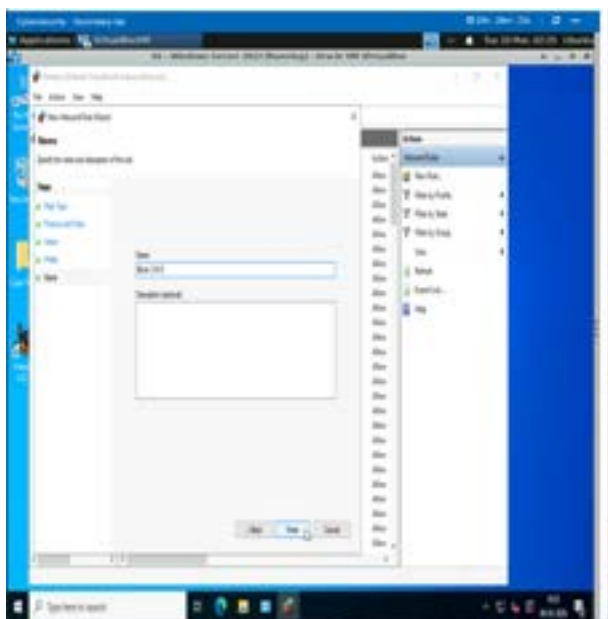
Select Block the connection → click Next



Check Domain, Private, and Public → click Next

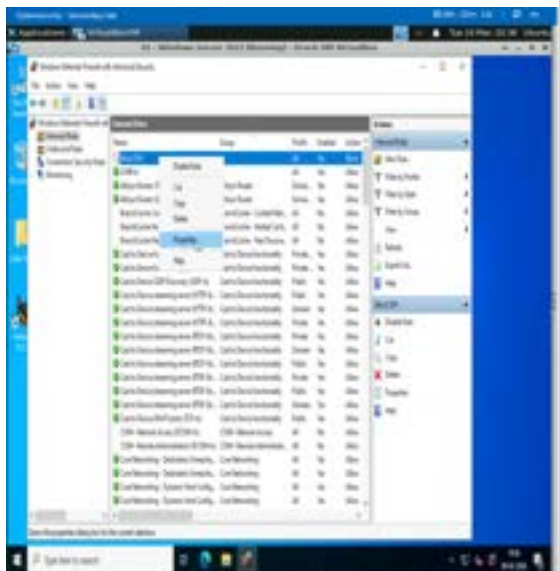


Give the rule a name like: Block SSH from Specific IP then click the Finish button



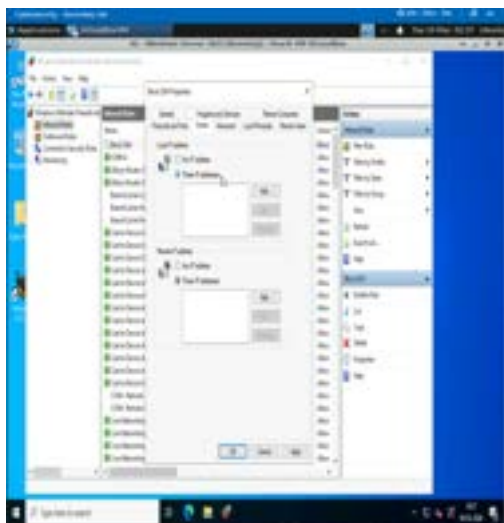
Step 4: Apply to a Specific IP Address

Right-click your new rule → select Properties



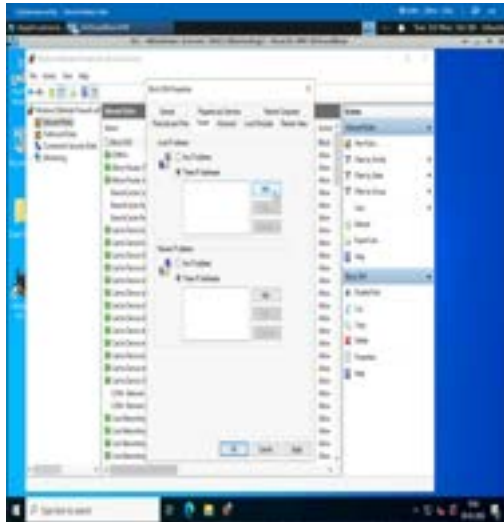
Go to the Scope tab then

Under Remote IP address, select These IP addresses

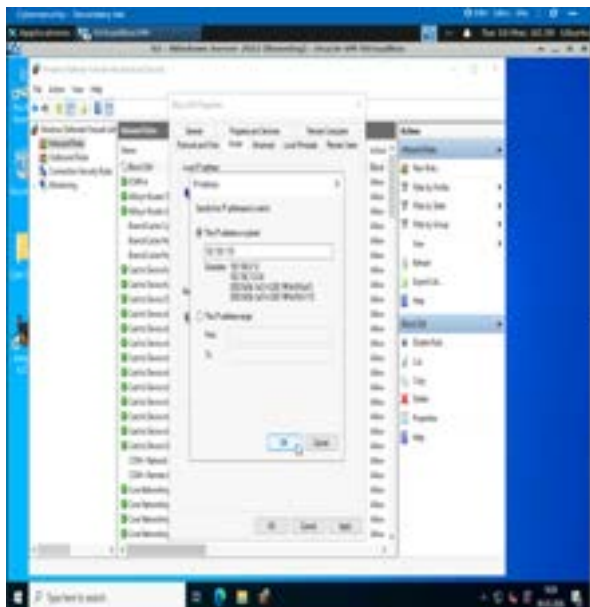


Click Add... and enter the IP address you want to block (for example, 192.168.1.50)

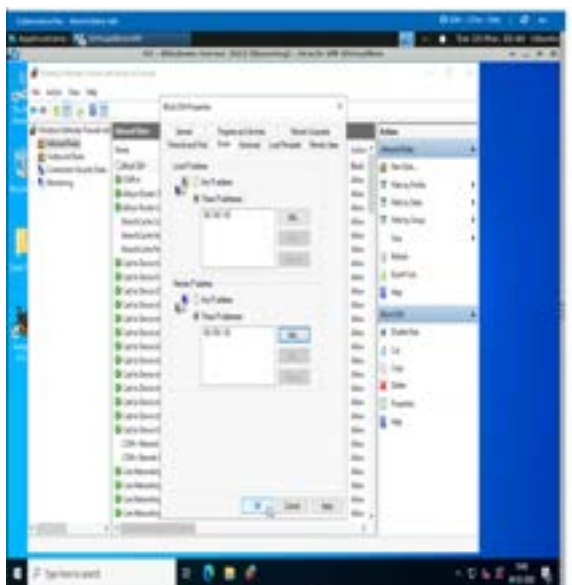
Click Add



Enter the IP address you want to block (for example, 192.168.1.50)

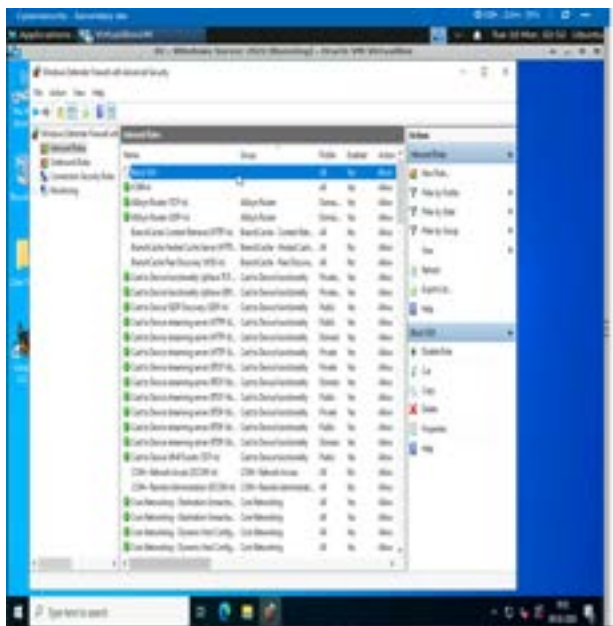


Click OK to apply

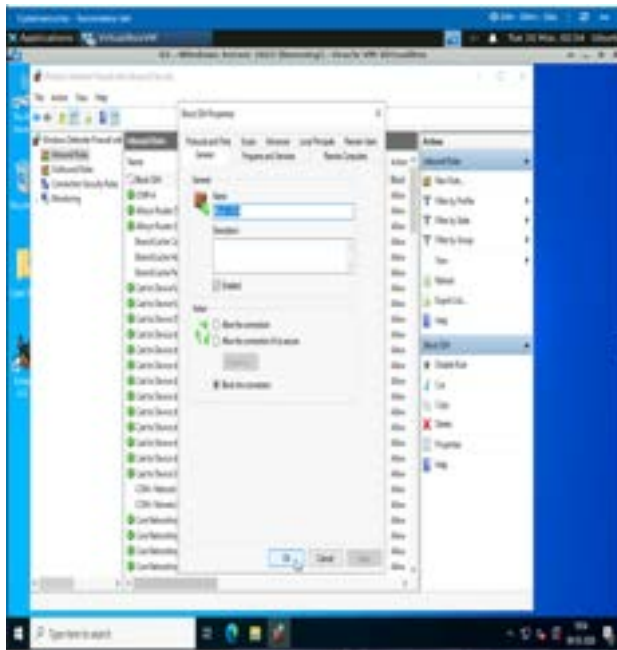


Step 5: Verify the Rule

. Make sure your rule is listed at the top of Inbound Rules



Try connecting via SSH from the blocked IP — it should now fail



Once you Click OK your Block SSH should be red at the top of the page.

